

SESSION 0

The Macro Toolkit

Five things the rest of this course assumes you have. One three-hour session, in class, before we open Chapter 1.

International Macroeconomics of the Euro Area · CY Cergy Paris Université

No prior macroeconomics course is assumed. Bring a laptop or a phone: three of the five blocks end with live data.

Why we start here

This course is built around one model, extended step by step.

What Session 1 assumes

- Output can be split into who spent it
- Saving is output not consumed
- A deficit is not a debt
- A euro next year is worth less than a euro now
- A price can be quoted two ways round

What it does not assume

- Any calculus
- Any prior model of the open economy
- Any familiarity with the euro crisis
- Any statistics beyond reading a chart

Every model in this course is drawn before it is written down.

The gap this session closes is vocabulary, not mathematics.

Five tools, and where each one is spent

#	Tool	What it lets you do	First needed
1	National accounting identities	Read a current account as saving minus investment	Session 1, Ch. 1–2
2	Stocks and flows	Tell a deficit from a debt, a current account from a NIIP	Session 1, Ch. 2
3	Real versus nominal	Separate price changes from quantity changes	Session 3, Ch. 9
4	Discounting and present value	Write down what a country can afford over time	Session 1, Ch. 3
5	Reading an exchange rate	Say which way the euro moved, and why inside the union it cannot	Session 3, then 5 and 6

Each block runs the same way: the idea, a worked example with numbers, a live data exercise you do yourself, and the mistake that block is designed to prevent.

Twenty to thirty-five minutes per tool across one three-hour session, plus the check at the end. Running order on the next slide.

TOOL 1 OF 5

The national accounting identities

Almost everything in this course is one identity, rearranged. Learn it once here and you will recognise it in every chapter.

Start with who bought the output

$$Y = C + I + G + NX$$

A country produces a quantity of goods and services in a year. Somebody bought every unit of it. This equation just lists the buyers. It is an identity, not a theory: it is true by construction, and no assumption can make it false.

C

Households, on consumption

I

Firms, on machines,
buildings, inventories

G

Government, on goods and
services

NX

Foreigners, on exports,
minus what we bought from
them

NX is also written TB, the trade balance. The two labels mean the same thing.

Rearrangement one: produce versus absorb

Call domestic spending **absorption**:

$$A = C + I + G$$

Then the identity becomes:

$$NX = Y - A$$

A trade deficit is not a sign that a country is losing money, and not a sign that its firms are uncompetitive. It is an arithmetic statement: **the country used more goods this year than it made.**

The rest of the world supplied the difference, and it did not do so for free. It lent the country the resources. That loan is the financial account.

So a deficit raises one question and only one: **was the borrowing worth it?** Sessions 2 and 3 answer it.

Rearrangement two: the spine of the course

STEP 1

$$S = Y - C - G$$

National saving is output not eaten by households or government.

STEP 2

$$Y - C - G = I + NX$$

Take C and G to the other side of the first identity.

STEP 3

$$CA = S - I$$

The current account is the gap between what a country saves and what it invests at home.

Read it in words. A country that saves more than it can usefully invest at home lends the surplus abroad, and runs a current-account surplus. A country that wants to invest more than it saves borrows from abroad, and runs a deficit. Nothing else is going on.

One refinement. CA and NX are close but not identical. The current account also includes net income earned on assets held abroad and net transfers. For a country with large foreign assets or liabilities, that gap is large, and Ireland is the European case where it is enormous.

Write $CA = S - I$ on the first page of your notes. You will use it in every session.

Worked example: two routes, one number

A country in one year, in billions of euro:

Output Y	100
Consumption C	60
Government G	22
Investment I	25

Assume no net income from abroad, so $CA = NX$.

Route A – spending side

$$NX = Y - (C + I + G) = 100 - 107 = -7$$

Route B – saving side

$$S = 100 - 60 - 22 = 18, \text{ so } S - I = 18 - 25 = -7$$

The country absorbed 107 and produced 100. It saved 18 and invested 25. Both sentences describe the same 7, borrowed from abroad.

Check yourself: if G rises to 27 and nothing else changes, what happens to S and to the current account?

Three traps in Tool 1

“A trade deficit means money is leaving the country.”

No. Goods come in and claims go out. The country receives more goods than it ships, and hands over IOUs. Money as such is not the object.

“Saving means money in a bank account.”

No. National saving is output not consumed. It is measured in goods, not in deposits. A country can have high saving and a shrinking money supply.

“GDP is the country's income.”

Not quite. GDP is output produced inside the borders. Some of it is income for foreign owners. GNI subtracts that. In Ireland the two differ by a wide margin.

The third trap is the one that matters most for a euro-area course. When profits of foreign-owned firms are booked in a country's GDP and then paid out abroad, the trade balance and the current account part company. You will see this again in Session 2.

Check for yourself later: compare Irish GDP with Irish GNI in the Eurostat table on the next slide.

Make the identity close, with real numbers

Core

1. Open the Eurostat table **nama_10_gdp**.
2. Choose France, then Spain. Unit: current prices, million euro.
3. Select B1GQ (GDP), P3 (final consumption), P5G (gross capital formation), P6 (exports), P7 (imports).
4. For one year, check that $P3 + P5G + P6 - P7$ equals B1GQ.
5. Compute $S - I$ and say whether the country lent or borrowed that year.

Extension

- Download the CSV and repeat for every year from 1999. Plot NX as a share of GDP.
- Add Ireland. Pull B1GQ and B5GQ (gross national income) and compare them.
- Open the Eurostat current-account table **teibp010** and check how far CA sits from NX for each country.

If the numbers do not close exactly, check that every series is in the same unit and the same price basis before you look for an economic reason.

TOOL 2 OF 5

Stocks and flows

Most confused arguments about European debt are a stock being mistaken for a flow. This block is twenty-two minutes of insurance against that.

A flow has a “per”. A stock has a date.

The test is a sentence. If the number only makes sense with the words *per year* or *per quarter*, it is a flow. If it only makes sense with the words *as of 31 December*, it is a stock. Every pair below is a flow feeding a stock.

Flow (per period)	Stock (at a date)	Where it turns up
Investment	Capital stock	Session 2, Ch. 5
Saving	Wealth	Session 1, Ch. 3
Current account	Net international investment position	Session 1, Ch. 2
Government deficit	Government debt	Session 2, Ch. 8
Net capital inflow	External debt	Sessions 4 and 6

A bath fills at a rate, in litres per minute. It holds a level, in litres. Turning the tap down does not empty the bath.

The one stock-flow relation you must know

$$\text{NIIP this year} = \text{NIIP last year} + \text{current account} + \text{valuation changes}$$

The net international investment position is what a country owns abroad minus what it owes abroad, at a date.

The flow term

A surplus this year adds to the position.
A deficit subtracts from it. This is the channel you model in Chapters 2 and 3.

The valuation term

Assets you already hold change price.
Nothing crosses a border, and the stock still moves. Exchange rates and equity prices do this constantly.

Why it matters

“Sustainability” in Session 1 is a statement about the stock. Policy talk is almost always about the flow. Keep the two apart.

In Chapter 2 the same relation is written with symbols and a no-Ponzi condition attached. The content is this line.

Worked example: the deficit falls, the debt grows

A country starts with a NIIP of -50% of GDP. It runs current-account deficits of 5% of GDP for three years. Then, under pressure, it cuts the deficit to 2% of GDP and holds it there. Ignore valuation changes.

Year	Current account	NIIP, end of year	What a headline says
0	—	-50	—
1	-5	-55	Deficit steady
2	-5	-60	Deficit steady
3	-5	-65	Deficit steady
4	-2	-67	Deficit cut by 60%
5	-2	-69	Adjustment working

The flow improved sharply. The stock got worse every single year. To stop the stock deteriorating the flow has to reach zero; to repair it the flow has to turn positive and stay there for a long time.

Check yourself: from year 5, with a surplus of 2% a year, how long to return to -50 ?

Watch a stock and its flow move apart

Core

1. Open the Eurostat table **tipsii10**, net international investment position, annual, as a share of GDP.
2. Select Spain, Portugal, Greece and Germany from 1999.
3. Find the year each country's position stopped getting worse.
4. Now open **teibp010** and find the year each country's current account turned positive. Compare the two dates.

Extension

- Download both series for one country. Cumulate the current account from 1999 and plot it against the actual NIIP.
- The two lines will not coincide. The gap is the valuation term. Say which direction it moved and guess why.

Do not look for an arithmetic mistake. The gap is real and it is one of the more interesting numbers in the dataset.

TOOL 3 OF 5

Real and nominal

Every important variable in this course comes in two versions. Interest rates, exchange rates, output, wages. Session 3 turns entirely on the difference.

Nominal is in euros. Real is in goods.

$$\text{real} = \text{nominal} \div \text{price index}$$

$$\text{real growth} \approx \text{nominal growth} - \text{inflation}$$

A nominal figure answers “how many euros?”. A real figure answers “how much stuff?”. The second is almost always the one you want, because the first moves whenever the price level moves, even if nothing has happened.

Dividing by a price index is called **deflating**. The index has to be stated: the same nominal series deflated by different indices gives different real series, and they can point different ways.

When Eurostat offers you “current prices” and “chain linked volumes”, it is offering you nominal and real.

Never compare a nominal figure across years or countries without saying which index you would deflate it by.

The real interest rate

You lend a euro and get back $1 + i$ euros. But prices rose by π , so those euros buy less. What you actually earned, measured in goods, is the real interest rate r .

Exact

$$1 + r = (1 + i) \div (1 + \pi)$$

Always right. Use it when inflation is high.

Approximate

$$r \approx i - \pi$$

Fine when both are small. Nearly all euro-area work uses this.

Which π ?

Ex ante uses expected inflation. It is what borrowers and lenders decide on, and it is unobservable.

Ex post uses realised inflation. It is what they got, and it is in the data.

In this course, r without further comment is a real rate. When a chapter says “the world interest rate”, it means a real one.

The ex ante and ex post gap is one reason unexpected inflation redistributes between debtors and creditors.

Worked example: when the shortcut breaks

Case A — a normal euro-area year

Nominal rate $i = 3.0\%$, inflation $\pi = 2.2\%$.

Shortcut: $r \approx 3.0 - 2.2 = \mathbf{0.80\%}$

Exact: $1.030 \div 1.022 - 1 = \mathbf{0.78\%}$

The difference is two hundredths of a percentage point.
Nobody will ever care.

Case B — a currency crisis

Nominal rate $i = 60\%$, inflation $\pi = 50\%$.

Shortcut: $r \approx 60 - 50 = \mathbf{10.0\%}$

Exact: $1.60 \div 1.50 - 1 = \mathbf{6.7\%}$

The shortcut overstates the real return by half. Here it matters a great deal.

Rule for this course: use the shortcut freely inside the euro area, and reach for the exact formula the moment you are looking at a country in the middle of a crisis. Sessions 5 and 6 are about countries in the middle of a crisis.

The same warning applies to the approximation for the real exchange rate.

LOOK AHEAD — SESSION 3

The same distinction, applied to a currency

Nominal exchange rate

E

The price of one currency in terms of another. A number you can read off a screen.

Real exchange rate

$$e = E \times P^* \div P$$

How many foreign baskets of goods one domestic basket buys. This is the one that governs trade.

Inside the euro

E = 1, frozen

So if e has to move, only P and P^* can move it. That means prices and wages.

You do not need to understand this yet. You need to have seen it once, so that when Session 3 arrives the idea is familiar rather than new. Chapters 9 and 10 do the work; this slide only plants the shape of the argument.

Compute a real interest rate yourself

Core

1. Open the ECB Data Portal inflation page and read the latest euro-area HICP annual rate.
2. Open the ECB key interest rates page and read the deposit facility rate.
3. Subtract. You now have an ex post real policy rate.
4. Say in one sentence what it means for someone holding cash.

Extension

- Download the monthly HICP series and the policy rate from 1999. Plot the difference.
- Mark the periods where it was negative. There are more of them than students expect.
- Then pull Eurostat GDP for Spain twice: current prices, and chain linked volumes. Compare 1999 to the latest year under each.

The second exercise is the nominal-versus-real trap in a single screenshot.

TOOL 4 OF 5

Discounting and present value

This is the block that makes Chapter 3 readable. By the end of it you will have written down the intertemporal budget constraint without being told that is what you were doing.

A euro next year is not a euro

Lend a euro today at rate r and next year you hold $1 + r$ euros. Run that backwards: to have one euro next year you need only $1 \div (1 + r)$ euros today.

So **$1 \div (1 + r)$ is a price**: the price of a euro next year, quoted in euros today. Discounting is nothing more mysterious than converting at that price, the way you would convert dollars into euros.

You may never add euros from different dates without converting first. That rule alone removes most of the confusion in Chapter 3.

Present value of an amount X received next year:

$$\text{PV} = X \div (1 + r)$$

At $r = 10\%$, a promise of 110 next year is worth exactly 100 today.

At $r = 5\%$, the same promise is worth about 104.8.

Higher interest rate, lower present value. This is also why a bond's price falls when its yield rises.

You have just written Chapter 3

$$C_1 + C_2 \div (1 + r) = Y_1 + Y_2 \div (1 + r)$$

The present value of what you consume over your life equals the present value of what you earn over your life.

It is an ordinary budget line

Two goods, one budget. The goods are consumption now and consumption later. The relative price is $1 + r$.

Borrowing is buying now

Consuming more than income in period 1 means moving along the line towards the present and paying $1 + r$ for each unit moved.

A country is a household

Replace “person” with “country” and “borrowing” with “current-account deficit”. That is the intertemporal theory of the current account.

Chapter 3 adds preferences and an optimum on top of this line. The line itself is what you have here.

Worked example: a country that borrows, then repays

A country lives two periods. It produces 100 this period and 110 next. It starts with no foreign assets or debts. The world real interest rate is 10%. It decides to consume 120 now.

Quantity	Period 1	Period 2	How you get it
Output Y	100	110	Given
Consumption C	120	88	Chosen, then forced
Trade balance Y – C	–20	+22	Subtract
Net foreign assets, end	–20	0	Cumulate
Current account	–20	+20	Trade balance plus interest

The constraint holds

$120 + 88 \div 1.1 = 200$, and $100 + 110 \div 1.1 = 200$. The country could not have consumed 120 and then 90.

Note the last two rows

The trade surplus is 22 but the current account surplus is only 20. The missing 2 is the interest paid on last period's borrowing.

A deficit today is a commitment to a surplus later. That sentence is the intertemporal theory of the current account.

Two frightening words, defused

“No-Ponzi condition”

In words: you cannot plan to pay interest by borrowing more, forever. Nobody lends against a plan that never repays anything.

So a country may run deficits for a long time, but the present value of what it eventually gives up has to cover what it borrowed.

“Transversality”

In words: at the end, you do not leave value on the table and you do not die owing. The discounted terminal position goes to zero.

In the two-period model this is simply the statement that assets at the end of period 2 are zero. It is the last row of the table you just filled in.

This is why “sustainability” is a real question rather than a slogan. A country's external position is sustainable if its expected future trade surpluses have a present value at least as large as what it owes. That is checkable. It is also, in Session 6, exactly what markets stop believing during a crisis.

Session 1 writes both conditions in symbols. You now know what they say.

Discounting you can see: bond yields

Core

1. Open the ECB Data Portal and find long-term government bond yields, ten years.
2. Read today's figure for Germany and for Italy.
3. Both are promises of euros on fixed future dates. The one with the higher yield has the lower present value per euro promised.
4. Say in one sentence why a lender would demand more from one than from the other.

Extension

- Download both series from 1999 and plot the Italian yield minus the German yield.
- Describe the shape in three phases without yet explaining it.
- Keep the file. Session 4 uses exactly this chart, and asks you to read it as an interest parity condition that failed.

Two countries, one currency, one central bank, and two different prices for the same promise. Notice that this is strange.

TOOL 5 OF 5

Reading an exchange rate

Always ask: what is in the denominator?

An exchange rate is a price, and a price is always “so much of this per one of that”. Two sources can quote the same market in opposite directions, and both are correct. The chart only means something once you know which way round it is.

The ECB convention

1 EUR = 1.08 USD

Foreign currency per one euro. When the number rises, the euro buys more dollars, so the euro is stronger.

This is how every ECB reference rate and chart is published.

The common textbook convention

E = domestic per foreign

Many textbooks define E as units of home currency per unit of foreign currency. When that number rises, the home currency is **weaker**.

Check the definition on the first page of each chapter. Do not assume it matches the last one.

The two conventions are reciprocals, so a rise in one is a fall in the other. Write the convention you are using at the top of every exercise you hand in. It costs one line and it saves marks.

The 1.08 figure above is an illustration, not today's rate. Read the live rate on the portal.

Four words that are not synonyms

Word	What happens	Who does it	Which regime
Appreciation	The currency buys more foreign currency	The market	Floating
Depreciation	The currency buys less foreign currency	The market	Floating
Revaluation	The official parity is moved up	The authorities	Fixed or pegged
Devaluation	The official parity is moved down	The authorities	Fixed or pegged

The distinction is not pedantry. Sessions 5 and 6 turn on it. A price that moves on its own and a parity that somebody has to decide to move are politically and economically different objects, and the second one can be defended, attacked, and lost.

Britain in 1992 did not experience a depreciation of sterling within the ERM. It left the mechanism. Greece after 2010 could not devalue, because there was no parity to move and no currency to move it with.

Keep this table. Session 6 is a catalogue of what happens when a parity cannot be held.

Bilateral, effective, real

Bilateral

One currency against one other. Easy to read, and usually the wrong thing to look at, because a country trades with many partners.

Effective

An index against a basket of partners, weighted by how much you trade with each. The euro can rise against the dollar and fall on an effective basis in the same week.

Real

Adjusted for the price levels on each side. This is the one that decides whether your goods are cheap or expensive abroad, and it is the subject of Session 3.

The crudest real exchange rate there is

Take one good sold in nearly every country. Compare its local price converted into euros with its euro-area price. If it costs more abroad, the euro is cheap in real terms against that country.

That is the whole logic of the Big Mac index. It is a joke with a serious point: one good is a terrible basket, but the arithmetic is the arithmetic of purchasing power parity, which is Chapter 9.

The Economist publishes the underlying data openly. The link is on the last slide.

Real appreciation means your goods got expensive relative to foreign goods, whatever caused it.

THE SENTENCE THE REST OF THE COURSE IS ABOUT

Between France and Italy the nominal exchange rate is 1. It was 1 yesterday. It will be 1 tomorrow.

So if the real exchange rate has to move, prices and wages are the only things left that can move it.

A floating economy carries the adjustment on its currency. A euro member carries it on its labour market. That is the whole bargain.

LIVE DATA — 10 MINUTES

Read a rate, then say which way it moved

Core

1. Open the ECB euro reference rates page. Read today's euro-dollar rate and state the convention out loud.
2. Open the ECB chart for the same rate over the last ten years.
3. Pick any two dates and say, in a full sentence, whether the euro appreciated or depreciated between them.
4. Now say what the same movement looks like under the opposite convention.

Extension

- Open the Economist's Big Mac data repository and download the raw file.
- Take the euro area and one non-euro European country. Compute the implied rate and compare it with the market rate.
- Try the ECB currency converter on a historical date and check your arithmetic against it.

Notice that the file carries one entry for the whole euro area, not one per member. Ask yourself why that is allowed, and you have asked the question Session 3 answers.

What the checks actually ask

The toolkit check at the end of today, and the diagnostic at the start of Session 1, both work through the same five questions. If you can answer these in your own words, you are ready for Chapter 1.

Tool	The question you must be able to answer
Identities	A country's saving is 18 and its investment is 25. Is it lending to or borrowing from the rest of the world, and by how much?
Stocks and flows	A country halves its current-account deficit. Does its net foreign debt fall? Explain in one sentence.
Real and nominal	The nominal rate is 4% and inflation is 3%. What is the real rate, and what does it mean for a saver?
Discounting	At $r = 10\%$, is a promise of 105 next year worth more or less than 100 today?
Exchange rates	The euro goes from 1.10 to 1.05 dollars. Did it appreciate or depreciate? Which convention were you just given?

Miss more than two and you should redo the relevant block before Session 2. Everything you need is in this deck, and I would much rather you came to see me in week one than in week five.

The Session 1 diagnostic is not graded. It exists so I know which of these five to slow down on.

Every link, in one place

Output and external position

[Eurostat nama_10_gdp](#) — GDP and its components

[Eurostat teibp010](#) — current account

[Eurostat tipsii10](#) — net international investment position

[Statistics Explained](#) — balance of payments, in words

Prices and interest rates

[ECB Data Portal](#) — the home page, start here

[ECB inflation](#) — euro-area HICP

[ECB rates](#) — policy rates and exchange rates

[ECB explainers](#) — short, neutral, primary source

Exchange rates and beyond

[Euro reference rates](#) — the daily fixing

[ECB currency converter](#) — convert at a past date

[Big Mac data](#) — the Economist's open repository

[Euro Crisis Monitor](#) — TARGET2 balances, used from Session 1

All of these are free and public. Every figure in this course can be reproduced from them.

What you can now do

Tool 1

Read a current account as saving minus investment, and refuse to moralise about a deficit.

Tool 2

Keep a deficit and a debt apart, and know that closing one does not undo the other.

Tool 3

Separate a price change from a quantity change, in output, in interest rates, and soon in currencies.

Tool 4

Write down a two-period budget constraint and say what no-Ponzi and transversality mean.

Tool 5

State a quote's convention before reading it, and tell a depreciation from a devaluation.

Next session we open the balance of payments, and ask why German claims on the Eurosystem rose while Greek and Spanish liabilities grew. You now have every tool that question needs.